Name: **KAUST ID:**

Assessment Form for HW2

Functionality	Subtasks	Points	Note
Data Parsing	Classification File Parse (Iris and Titanic)	2	
	Show the statistical features	1	
Basic function	Gaussian Basis	3	
	Sigmoidal Basis	3	
	Interface to choose which basis	1	
Bayesian Linear Regression	Model Implementation	12	
	Test on the synthetic and real datasets(at least two datasets among Automobile and Forest, Wine**)** (You need to plot corresponding figures for them).	**10**	
	Plot 95% confidence intervals for at least 2D data and analysis.	**3**	
	Output MSE, plot predictions over data samples, and analysis over three basis functions.	**10**	
	Plot and output posterior parameters for at least 2D data and analysis.	**2**	
Linear Classification	Implement logistic regression	10	
	Implement Gaussian Discriminant Analysis	10	
	Experiments on synthetic and real classification datasets**(**Iris and Titanic**)** (You need to plot corresponding figures for them).	10	
	Plot the MSE over the iteration of Logistic Regression via gradient descent and plot predictions over data samples over three basis functions.	8	

	Evaluate and compare classification accuracy over three basis functions.	8	
	Evaluate and compare confusion matrices over three basis functions.	5	
	Evaluate and compare decision boundaries (at least for 2D synthetic data).	2	

Note：

1. For Gaussian Discriminant Analysis, you can have more information from Session 4.2 of Pattern Recognition and Machine Learning(Bishop,2006)

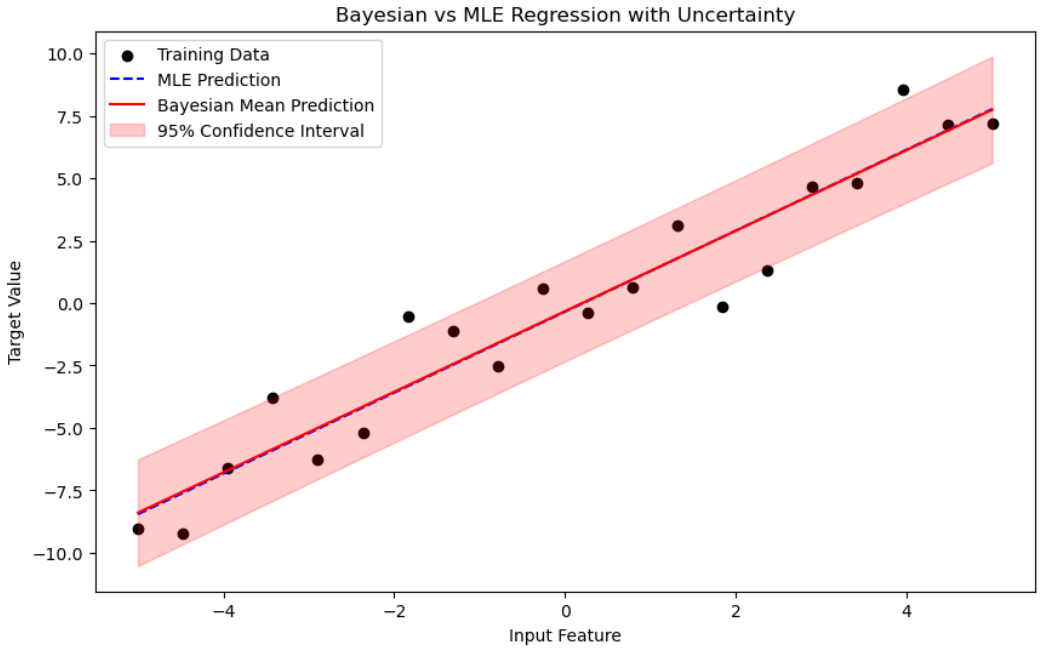
2. $Accuracy = \frac{Number\ of\ correctly\ classified\ samples}{Total\ Samples}$

3. For 95% Confidence Intervals(CI): $t = t_{pre} \pm 1.95 * \sigma, \sigma\ is\ the\ standard\ deviation\ of\ posterior$

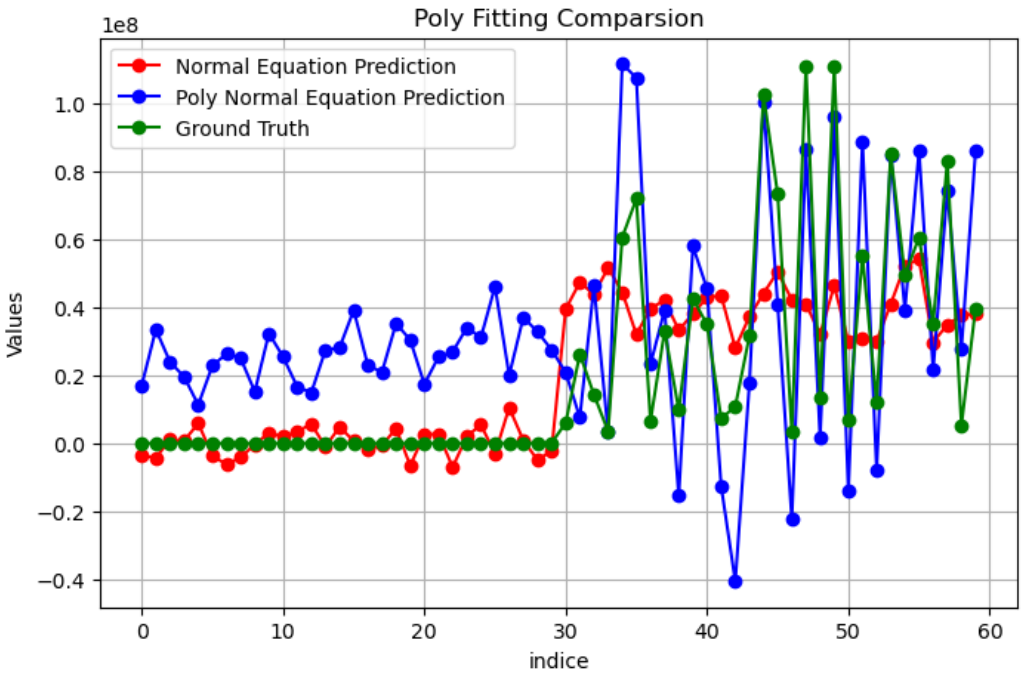
4. For **confidence intervals, posterior parameters, and decision boundaries**, you should at least plot the 2D synthetic data. For high-dimensional real data, it is not compulsory.
5. You can develop something outside of the requirements and you might gain some extra points according to the functionality complexity and your understanding.
6. The extra points can be used for making up the lost points of the compulsory points through all assignments.
7. You can have a maximum of 20 extra points for each assignment.

Plot Reference

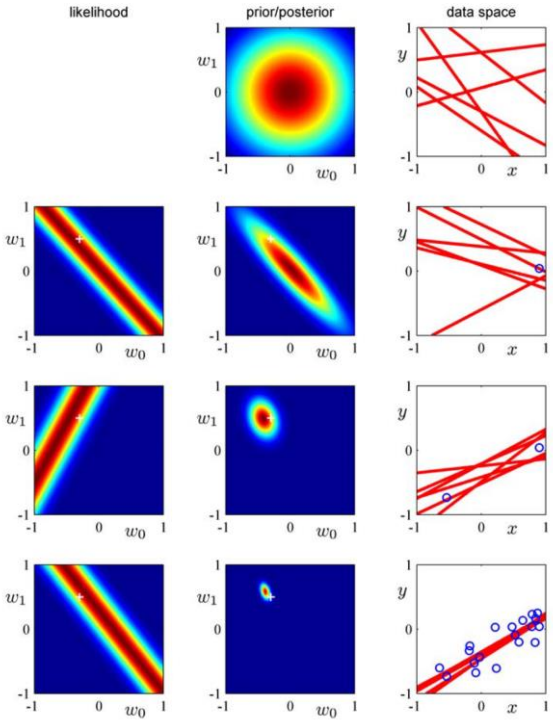
1. Confidence Intervals, I show you a 1D case, you should at least extend it to 2D.



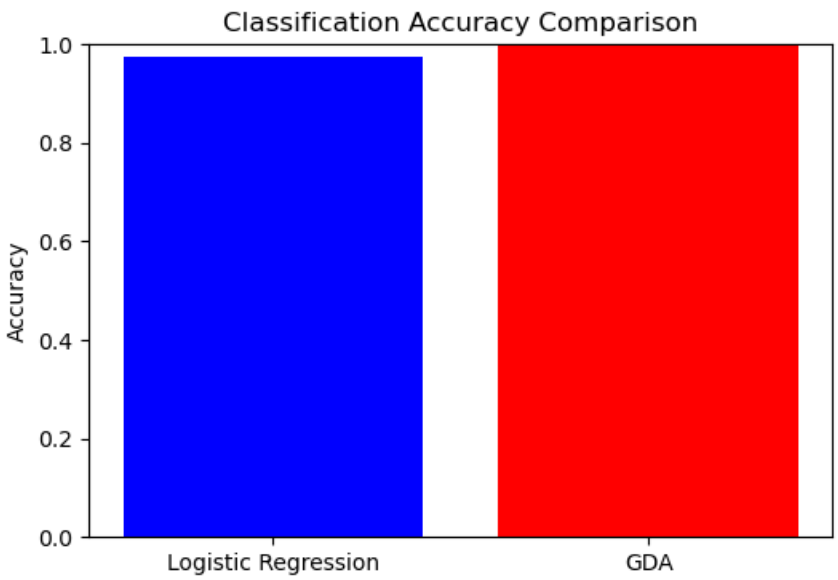
2. Predictions over data samples, x is the data sample indices, y is the numerical value



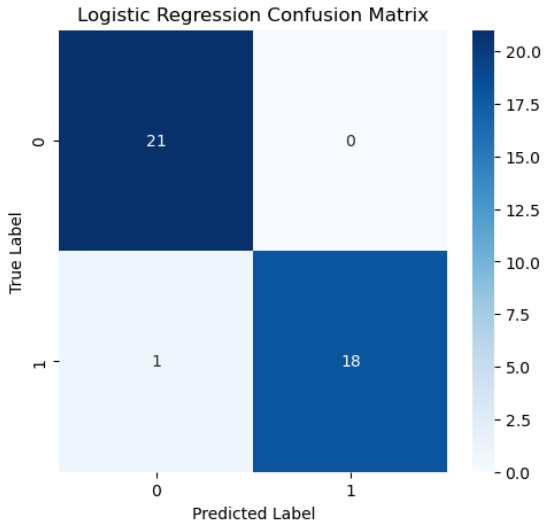
3. Posterior, for 2D case, you can check P155 of Pattern Recognition and Machine Learning(Bishop,2006). You can just show the final result instead of showing the learning progress.

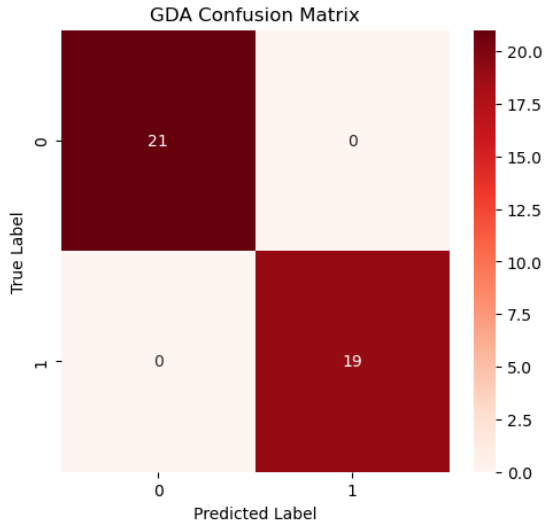
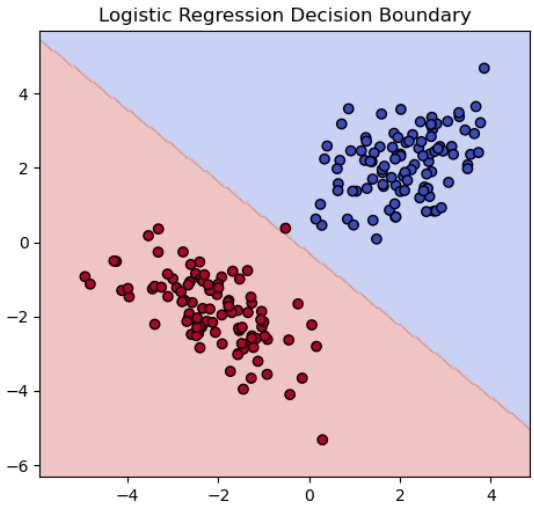
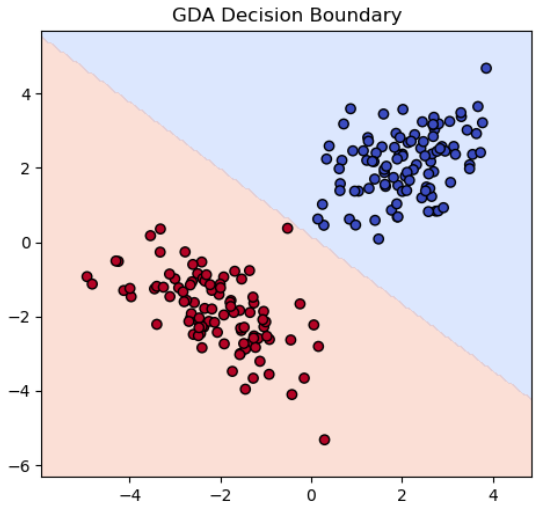


4. For classification accuracy.



5. For confusion matrices.





6. For decision boundaries(2D case)